



GE VERNOVA

ADMS

16.25.1

Communication Gateway Programmer Guide

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Change History

Date	Change Description
Jun 7, 2023	ADMS 3.16: • Updated Mobility Gateway Configuration. • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • Renamed document to ADMS Communication Gateway Programmer Guide. • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Apr 4, 2024	ADMS 16.7.0: • Updated document format and styling.
Sep 23, 2024	ADMS 16.12.0: • Added section Coded Translations. • Updated section Mobility Gateway Configuration. • Updated section Web Services Hosted by Mobile Enterprise (Inbound). • Added chapter Optional Features.
Nov 7, 2024	ADMS 16.14.0: • Documented how Mobile Enterprise updates crew shift and availability statues via Mobility Gateway.
June 25, 2025	ADMS 16.18.0: • Documented how to synchronize ADMS crew models with Mobile Enterprise via Mobility Gateway.

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1. Introduction

This document describes the ADMS Mobility Gateway used for integration with GE's Mobile Enterprise solution. The Mobility Gateway is a client (an initiator of interactions) of Mobile Enterprise REST endpoints and also a client of REST endpoints provided by OMS and DMS servers.

The Mobility Gateway interface uses HTTP connections with JSON messaging.

1.1. Overview

ADMS Mobility Gateway interfaces with GE's Mobile Enterprise (ME) system through the Mobile Enterprise S3 Adapter. Within ADMS, the Mobility Gateway interfaces with the DMS Interface server, Crew Server, Work Model Event Handler, OMS Interface Server, and Tagging Service, as shown in image below. Within ADMS, the DMS Interface server also communicates with the Switching Order server and Crew Server.

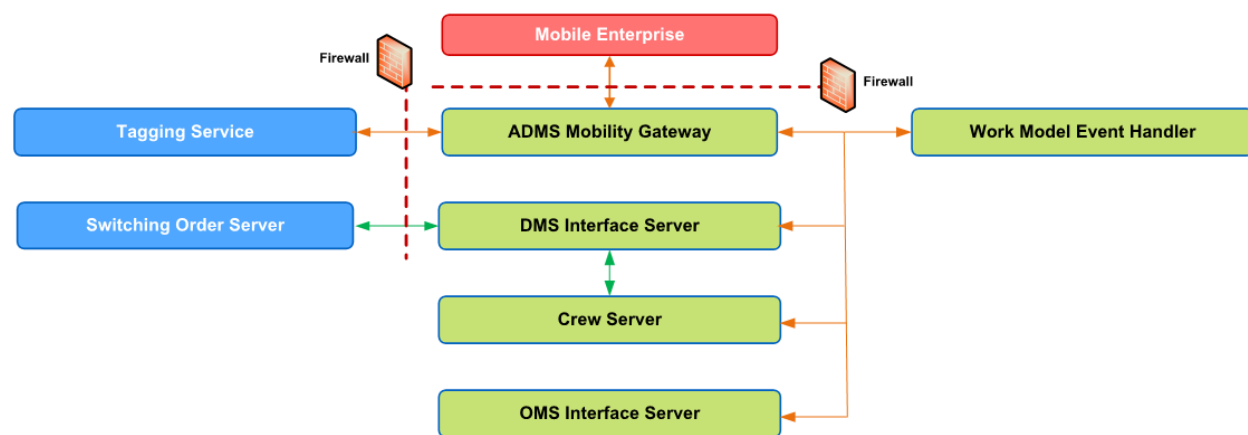


Figure 1. Mobile Enterprise Interface - Overview

The ADMS architecture provides the flexibility of running the Mobility Gateway, DMS Interface server, and other applications on different physical hosts. DMS Interface server communicates with Switching Order server via TCP/IP and with Mobility Gateway via HTTP.

1.2. High Availability

Mobility Gateway does not support high availability (redundancy).

1.3. Security

All messages between the ADMS Mobility Gateway and Mobile Enterprise can be exchanged using the HTTPS secure communications protocol (two-way SSL certificates for authentication).

To provide additional security, you can configure Mobile Enterprise to require an authorization header for all requests from ADMS. The authorization header, combined with Mobile Enterprise web service configuration parameters, is used by Mobile Enterprise to verify that requests come from a trusted client (ADMS).

Requests from Mobile Enterprise to ADMS do not need to include an authorization header. Mobile Enterprise sends messages to ADMS only in response to message requests from ADMS (polling). HTTPS ensures that the response from Mobile Enterprise (which contains requests for ADMS) is not tampered with during transmission.

The Mobility Gateway supports the User Account and Authentication (UAA) feature. For more information about setting up UAA, refer to the *ADMS Software Installation and Maintenance Guide*.

ADMS web API application servers also support using SSL certificates for the HTTPS mode.

1.4. Time Format

The time stamp format used by Mobility Gateway web interfaces is expressed as a string date and time value in the "Round Trip Format" (a string date and time value with time zone information).

For example:

- "2022-05-24T11:59:13.000-07:00" – this string is expressing a local time (-07:00 timezone), which corresponds to May 24, 2020 11:59:13 AM, Pacific Time
- "2022-05-24T18:59:13.000Z" – this string is expressing the same time instant in the UTC time zone ("Z").

1.5. Latitude/Longitude Format

The latitude and longitude used in messages throughout the different interfaces is expressed in decimal degrees. A coordinate converter is used to convert between the latitude/longitude and system x/y coordinates.

1.6. Coded Translations

The ADMS definition of Coded Translations dictates the values that users can select (for example, cause and sub-cause). When a field user selects values from Coded Translations, these values must match with those recognized by ADMS to ensure the message is properly communicated.

To ensure consistency between ADMS and Mobile Enterprise, Mobility Gateway reads the complete Coded Translations configuration and sends it to Mobile Enterprise.

There are two configuration settings that control the behavior of this integration:

- CodedTranslationXMLPath specifies the location of the CodedTranslationConfig.XML file in the

ADMS installation.

- CodedTranslationsUri specifies the Mobile Enterprise service endpoint to which the Coded Translation information is to be sent.

These settings are part of the *Mobile Enterprise Web Service Client* plugin definition in the *MobilityGatewayAppConfig.xml* configuration file.

Refer to [Mobile Enterprise Web Service Client Plugin Parameters](#) for more details.

Refer to [Web Services Hosted by Mobile Enterprise \(Inbound\)](#) for details on the **configuration/codedTranslations** endpoint.

Important

The ADMS Mobility Gateway sends the complete Coded Translations configuration to Mobile Enterprise only during startup. If any ADMS Coded Translation is changed, the ADMS Mobility Gateway must be restarted to send the updated values to Mobile Enterprise.

1.7. Crew Synchronization

Mobile Enterprise must always be aware of all crew model details in ADMS. At startup, Mobility Gateway sends Mobile Enterprise crew model details from the configured .crew model file, along with crew details from the journal files. When it is running, Mobility Gateway automatically synchronizes ADMS crew model details with Mobile Enterprise as they are updated.

Mobility Gateway automatically synchronizes crew model details with Mobile Enterprise anytime a crew is newly added, updated or deleted by:

- Crew Modeling Tool. Saved changes to .crew model file for any crew model addition, update or deletion.
- Crew Modeling API. Any journaled crew model addition, update or deletion.
- OMS Display - Crew Update Dialog. Any journaled crew model addition, update or deletion.

Important

Mobility Gateway will not synchronize crews with Mobile Enterprise for following journaled crew fields:

- Status
- ShiftStatus
- MobileCommunicationStatus
- AvailabilityStatus
- CrewWorkStatus
- CurrentLocation

- IncidentId
- IncidentName
- Active

Refer to [Mobile Enterprise Web Service Client Plugin Parameters](#) for more details about crew synchronization URL.

2. General Message Flow

This chapter provides an example scenario, which demonstrates the message flow between ADMS and Mobile Enterprise.

2.1. Simple Transformer Outage

This is a simple scenario where customers call the utility to report the lights are out. ADMS predicts an outage, and a Dispatcher sends a Crew to restore the power.

Several customers connected to the same transformer experience power loss and call the utility's Customer Service Representative, who raises a Trouble Ticket in the utility's Customer Information System. The ADMS Outage Management System analyses this information and predicts a blown transformer fuse. Next, OMS creates an incident to track the Unplanned Outage (with an outage status of *Predicted*). EOMS also creates a Restoration Order with an automatically created *Check Prediction* task to manage crew activities. Finally, the Incident and Restoration Order are packaged into a Trouble Order and sent to Mobile Enterprise, ready for dispatch to the field.

A Dispatcher allocates and dispatches a crew to the Restoration Order within EOMS. Next, EOMS adds the crew to the Trouble Order within Mobile Enterprise, which sends the order to the Mobile Enterprise application running on the Crew's mobile device.

The Crew arrives at the Trouble Order location, confirms that the transformer fuse is out, re-fuses the transformer, and restores all power. The Crew uses the Mobile Enterprise application to keep ADMS (and therefore the Dispatcher) informed of their progress while working on the Trouble Order. ADMS communicates the progress to external systems (such as the Customer Information System). Mobile Enterprise reporting includes accepting the order and traveling to the predicted transformer, confirming the outage location, and restoring power.

Once the Crew has restored power, they capture the final outage information using the Mobile Enterprise application, complete the order (on their device), and send this to ADMS for processing and viewing by the Dispatcher. ADMS sets the status of the Restoration Order to *Field Complete*. The Dispatcher then reviews the order history and makes any necessary amendments. Once satisfied, the Dispatcher closes the order in the EOMS. Mobile Enterprise then closes the Trouble Order.

Triggering event	Assumption
Multiple customer calls	The customer calls are a result of a blown fuse.

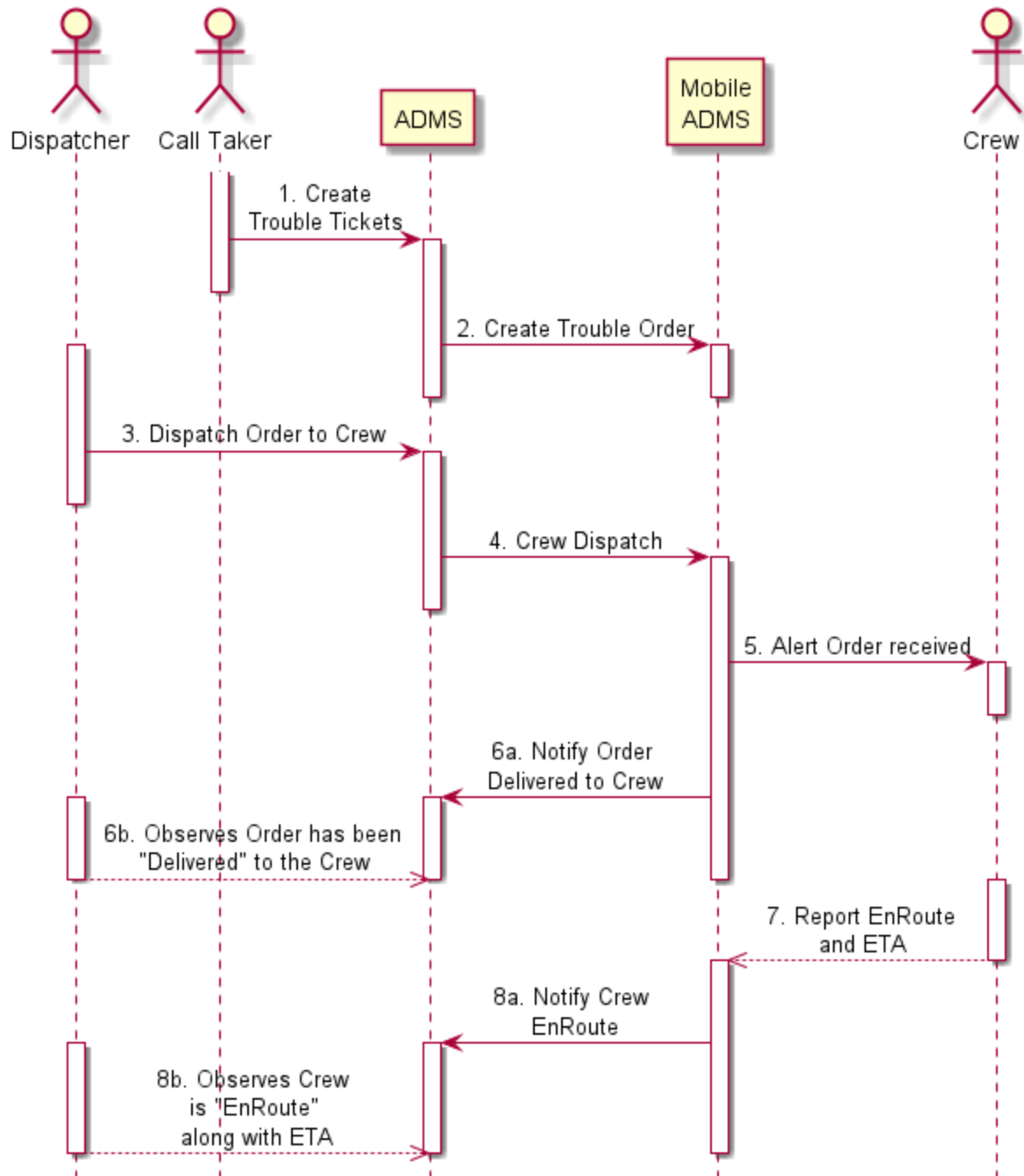
Step-by-step analysis of the scenario:

Summary

Scenario name	Primary actor	Triggering event	Pre-condition	Post condition
Creation and Dispatch	Customer Service	Customer Service representative creates a Trouble Ticket.	A fuse is blown, and customers call to report an outage.	ADMS creates a Trouble Order that includes the predicted Incident and the related Restoration Order.

Scenario name	Primary actor	Triggering event	Pre-condition	Post condition
	Dispatcher	The Dispatcher assigns and dispatches a Crew to the Restoration Order.	ADMS has created a Restoration Order.	The Crew receives the Trouble Order on the Field Client, and the Dispatcher can see that they are on the way.
Confirmation	Crew	The Crew confirms the predicted device.	The Crew has received the Trouble Order and arrived on site.	ADMS confirms the predicted device.
Restoration	Crew	The Crew restores power.	The repair is done.	All customers have their power back on.
Close Out	Dispatcher	The Dispatcher closes out the Restoration Order.	The outage is restored, and the Crew has returned their final outage information.	The Restoration Order is closed in ADMS.

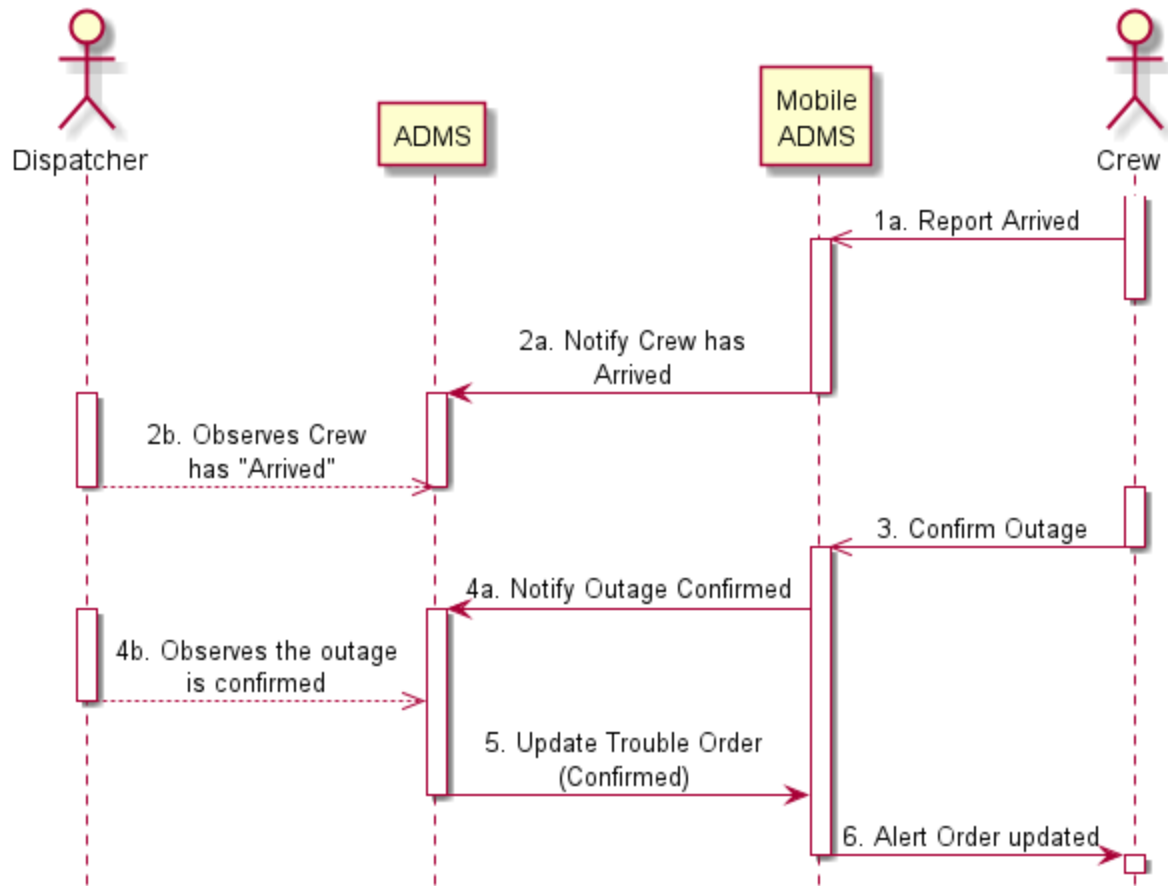
Creation and Dispatch



Step and Event		Process / Activity	
1	Customer Calls	Process	Trouble Call Entry
		Description	Customer calls the utility. Customer service representative creates a Trouble Ticket, which is sent to ADMS. ADMS creates a <i>Predicted</i> Incident and a Restoration Order with a <i>Check Prediction</i> task.
		Actors	Call Taker → ADMS
		Information Exchanged	Trouble Ticket
2	ADMS sends Trouble Order to Mobile Enterprise	Process	Order Sync
		Description	The known information regarding the unplanned outage is sent to Mobile Enterprise as a Trouble Order.
		Actors	ADMS → Mobile Enterprise
		Information Exchanged	Trouble Order
3	Dispatch Trouble Order to Crew	Process	Crew Dispatch
		Description	Dispatcher dispatches the Trouble Order to the Crew.
		Actors	Dispatcher → ADMS
		Information Exchanged	Assignment
4	ADMS assigns task	Process	Crew Dispatch
		Description	ADMS assigns the Crew to the <i>Check Prediction</i> task of the Restoration Order and an Update to the Order is sent to Mobile Enterprise.
		Actors	ADMS → Mobile Enterprise
		Information Exchanged	Trouble Order
5	ADMS request dispatch	Process	Crew Dispatch
		Description	ADMS sends a request to dispatch the Order to a Crew.
		Actors	ADMS → Mobile Enterprise
		Information Exchanged	Assignment
6	Crew alerted of a new Order	Process	Crew Dispatch
		Description	The Crew is alerted that there is a new Order.
		Actors	ADMS → Mobile Enterprise
		Information Exchanged	Assignment

Step and Event		Process / Activity	
7	Mobile Enterprise notifies Order delivered	Process	Crew Dispatch
		Description	Mobile Enterprise notifies ADMS of successful order delivery. The Dispatcher can see that the Trouble Order has been delivered to the Crew.
		Actors	Mobile Enterprise → ADMS
		Information Exchanged	Assignment Status
8	Crew goes enRoute and provides an ETA	Process	Field Crew Work
		Description	Crew goes enRoute and provides an ETA to the Order location.
		Actors	Field Client → Mobile Enterprise
		Information Exchanged	Status Update with ETA
9	Mobile Enterprise notifies crew assignment status (EnRoute) and ETA	Process	Field Crew Work
		Description	Mobile Enterprise notifies ADMS that the Crew is EnRoute along with an ETA. The Dispatcher can see that the Crew is EnRoute along with the ETA.
		Actors	Mobile Enterprise → ADMS
		Information Exchanged	Assignment Status with ETA

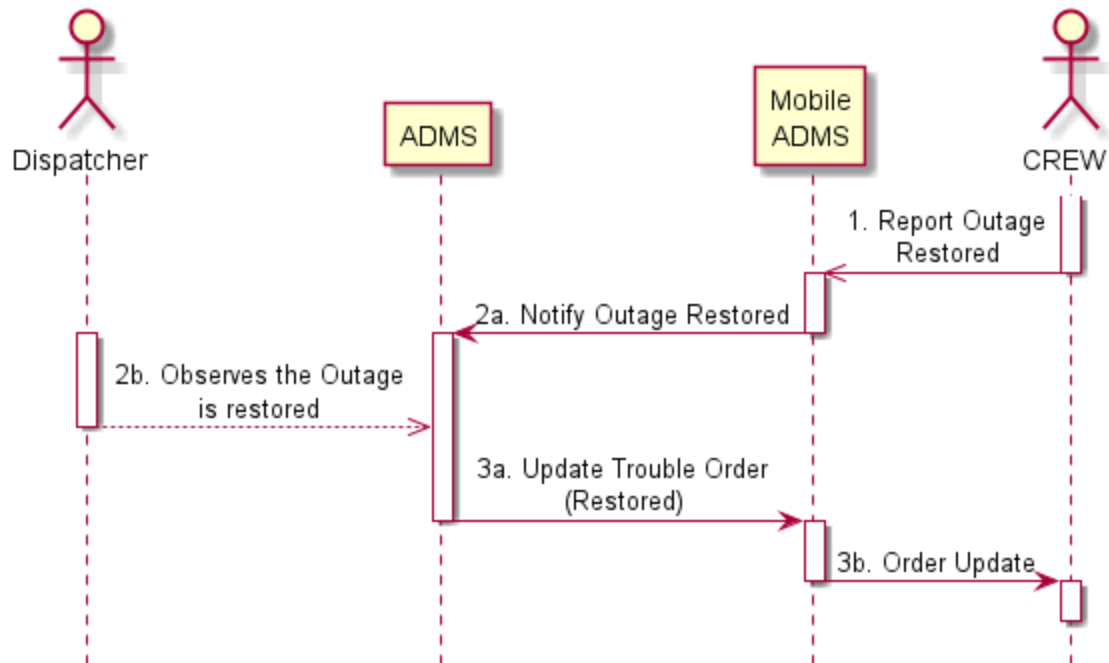
Confirmation



Step and Event		Process / Activity	
1	Crew arrives on site	Process	Field Crew Work
		Description	Crew arrives at the site location, confirms their arrival, and starts work.
		Actors	Crew → Mobile Enterprise
		Information Exchanged	Status Update
2	Mobile Enterprise notifies crew assignment status (Arrived)	Process	Field Crew Work
		Description	Mobile Enterprise notifies ADMS that the Crew has arrived. The Dispatcher can see that the Crew has arrived.
		Actors	Mobile Enterprise → ADMS
		Information Exchanged	Assignment Status

Step and Event		Process / Activity	
3	Crew confirms device open	Process	Fault Location Reporting
		Description	Crew identifies the transformer fuse is out and confirms the device is open.
		Actors	Crew → Mobile Enterprise
		Information Exchanged	Device Confirmation
4	Mobile Enterprise notifies device confirmed open	Process	Outage Management
		Description	Mobile Enterprise notifies ADMS that the device is confirmed open. ADMS updates the state of the device and the <i>Check Prediction</i> task against the Restoration order. ADMS also creates a new <i>Repair</i> task for the device. The Dispatcher can see that the device is confirmed open.
		Actors	Mobile Enterprise → ADMS
		Information Exchanged	Outage status (device confirmed)
5	ADMS sends order update to Mobile Enterprise	Process	Order Sync
		Description	ADMS sends an updated order which includes the updated state of the device and the new and updated tasks.
		Actors	ADMS → Mobile Enterprise
		Information Exchanged	Order
6	Mobile Enterprise sends order update to Field Client	Process	Order Sync
		Description	The Crew is alerted that the order has been updated.
		Actors	Mobile Enterprise → Crew
		Information Exchanged	Order

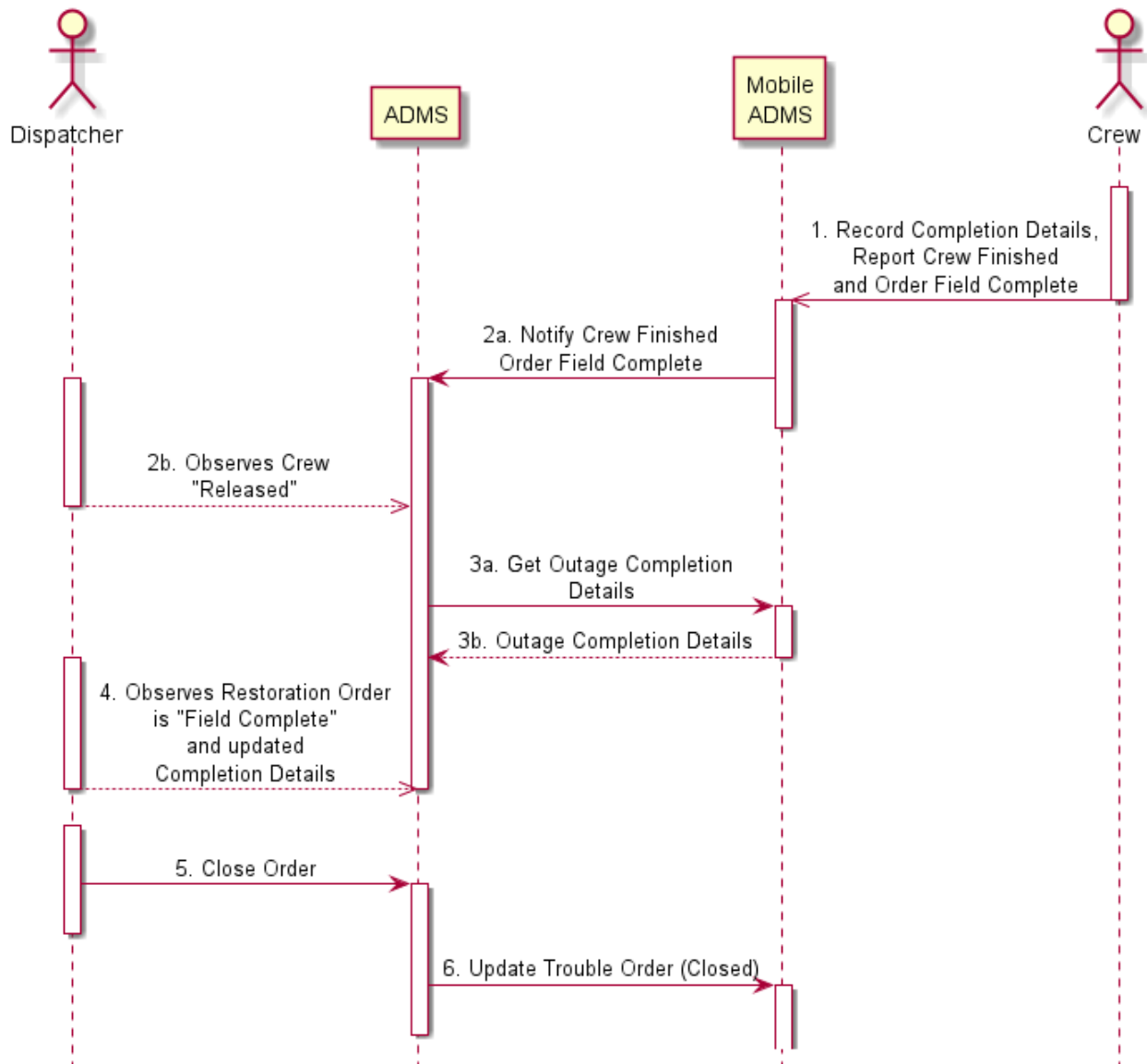
Restoration



Step and Event		Process / Activity	
1	Crew notifies that the outage is restored	Process	Field Crew work
		Description	The Crew repairs the transformer fuse and reports that all power has been restored.
		Actors	Field Client → Mobile Enterprise
		Information Exchanged	Outage Status (<i>Restored</i>)
2	Mobile Enterprise notifies ADMS that all power is restored	Process	Field Crew Work
		Description	Mobile Enterprise notifies ADMS that the Crew has restored power to all customers. ADMS updates the Incident outage status to <i>Restored</i> and completes the <i>Repair</i> task against the Restoration Order. The Dispatcher can see that the Incident outage status is restored.
		Actors	Mobile Enterprise → ADMS
		Information Exchanged	Outage status (<i>Restored</i>)

Step and Event		Process / Activity	
3	ADMS sends order update to Mobile Enterprise	Process	Order Sync
		Description	ADMS sends Mobile Enterprise an update of the order, which includes the following: • Outage status (<i>Restored</i>) • State of the device • Completed <i>Repair</i> task Mobile Enterprise then sends the updated order to the Crew's mobile device.
		Actors	ADMS → Mobile Enterprise → Field Client
		Information Exchanged	Updated Trouble Order

Close Out



Step and Event		Process / Activity	
1	Crew finishes Trouble Order	Process	Field Crew Work
		Description	Once all power has been restored, the Crew captures the outage completion details and finishes the Order. Typically, a single crew assigned to the outage fills out the outage completion details, which include the following: - Whether all crew have finished working on the outage (including all post-restoration activities) - Cause of the outage
		Actors	Field Client → Mobile Enterprise
		Information Exchanged	Status Update and Outage Completion details

Step and Event		Process / Activity	
2	Mobile Enterprise sends the Outage Completion notification to ADMS	Process	Field Crew Work
		Description	Mobile Enterprise notifies ADMS of the finish status for the crew. The Dispatcher can see that the Crew is <i>Released</i> on the Restoration Order.
		Actors	Mobile Enterprise → ADMS
		Information Exchanged	Crew Assignment
3	ADMS requests Outage Completion details from Mobile Enterprise	Process	Field Crew Work
		Description	ADMS requests Outage Completion details for the Trouble Order.
		Actors	Mobile Enterprise → ADMS
		Information Exchanged	Outage (completion details)
4	ADMS updates the Restoration Order to <i>Field Complete</i>	Process	Field Crew Work
		Description	ADMS updates the Restoration Order to <i>Field Complete</i> . The Dispatcher can see the updated status and the completion details.
		Actors	Mobile Enterprise → ADMS
		Information Exchanged	Restoration Order
5	Dispatcher closes the Order	Process	Order Close Out
		Description	The Dispatcher reviews the order history and makes any necessary amendments. Once satisfied, the Dispatcher closes the Restoration Order.
		Actors	Dispatcher → ADMS
		Information Exchanged	Updated outage information and Closed status
6	ADMS sends order update to Mobile Enterprise	Process	Order Close Out
		Description	ADMS sends an update of the order, including the outage status (<i>Closed</i>), to Mobile Enterprise.
		Actors	ADMS → Mobile Enterprise
		Information Exchanged	Updated Trouble Order

3. Interface Messages

The use of web services means that the Mobility Gateway does not maintain a constant connection with connected applications. Each request is a new connection. To restrict connections originating from less secure zones, all message exchanges are initiated by ADMS. ADMS sends web request messages to Mobile Enterprise or "pulls" messages from Mobile Enterprise via cyclic polling, to which Mobile Enterprise responds with queued messages.

3.1. Interface Characteristics

The following table represents inbound and outbound APIs provided by Mobility Gateway to facilitate integration with Mobile Enterprise:

Name	Technology	Purpose
Inbound API	REST/JSON	Mobility Gateway uses inbound APIs (web service) to receive switching order updates from the DMS Interface Server.
Outbound API	REST/JSON	Mobility Gateway uses outbound APIs (web client) to update data within Mobile Enterprise. Other outbound APIs manage switching orders, assignments, and crew resourcing. For more details, see Web Services Hosted by Mobile Enterprise (Inbound) .

3.2. Service Types

The ADMS Mobility Gateway web interface is configured in the `MobilityGatewayAppConfig.xml` configuration file (by default, stored at `%IDMS_ROOT%\config\interfaces`). Web services and web service clients are defined as separate plugins. For web services, you can configure the base URI. For web service clients, you can configure full URI endpoints.

The ADMS Mobility Gateway supports several type of plugins. The following list shows some of the plugins:

- **WebService:** This plugin implements a web service using the Microsoft Owin technology. All plugin definitions with the `WebService` type are services that are hosted by Mobility Gateway. These are internal endpoints / services, which are not meant for public usage.

The following example shows a `WebService` plugin definition:

```
<PluginDefinition>
  <Enabled>true</Enabled>
  <Type>WebService</Type>
  <Name>MobileSwitchOrderWebService</Name>
  <Config>
    <Name>MobileSwitchOrderWebService</Name>
    <Description>Implements Mobile Gateway Switch Order Plugin Web Service</Description>
    <BaseUri>http://+:8443/api/adms/MobileSwo</BaseUri>
```

In this example, the BaseUri parameter defines the base URI for the web service.

- **WebServiceClient:** This type implements a web service client interface, which is used to send requests to HTTP endpoints. This type is used to communicate with existing endpoints in the external Mobile Enterprise system and with internal services. The following example shows a WebServiceClient type plugin definition:

```
<PluginDefinition>
  <Enabled>true</Enabled>
  <Type>WebServiceClient</Type>
  <Name>MobileEnterpriseWebServiceClient</Name>
  <Config>
    <Name>MobileEnterpriseWebServiceClient</Name>
    <Description>Implements Mobile Enterprise Web Service Client</Description>
    <BaseUri>http://localhost:8445/api/ext/MobileEnterprise/v1</BaseUri>
    <OrdersUri>orders</OrdersUri>
```

In this example, the BaseUri parameter defines the base URI and the OrdersUri parameter defines the Orders endpoint. Therefore, the Orders URI will be formed as:

```
http://localhost:8445/api/ext/MobileEnterprise/v1/orders
```

- **LogService:** This type is used to define plugin logging settings:

```
<PluginDefinition>
  <Type>LogService</Type>
  <Name>MobilityGatewayLogService</Name>
  <Enabled>true</Enabled>
  ...
```

- **QueryService:** This type is used to communicate with existing endpoints of the OMS Interface Server query web service. The QueryService type functions similarly to the WebServiceClient type.

```
<PluginDefinition>
  <Enabled>true</Enabled>
  <Type>QueryService</Type>
  <Name>OmsInterfaceQueryService</Name>
  <Config>
    <BaseUri>http://localhost:18802/api/adms/oms/v2</BaseUri>
    <OmsQueryUri>Oms</OmsQueryUri>
```

3.3. Message Details

This section provides details for Mobility Gateway messages.

3.3.1. Web Services Hosted by Mobility Gateway (Inbound)

! Important

This document does not provide details for inbound messages, which are intended for internal usage.

Mobility Gateway uses inbound APIs to receive switching order updates from the DMS Interface Server and to receive events from the E-OMS Work Model.

3.3.2. Web Services Hosted by Mobile Enterprise (Inbound)

The following are default Mobile Enterprise web interface URLs that the Mobility Gateway communicates with. The Data Type field shows the data type that is accepted by Mobile Enterprise. For more details about the data type format (message schema), see the mobile-enterprise-order-web-service.json Swagger definition file (recommended for viewing using <https://editor.swagger.io/>) and the mobile-enterprise-order-web-service.html file. These files are delivered with the ADMS documentation.

- **/api/ext/MobileEnterprise/v1/**: This is the root of the Mobile Enterprise web service.

- **orders**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system creates an order.
Description	When an order is created in ADMS, Mobility Gateway sends order details to the Mobile Enterprise system. Mobile Enterprise creates the order within its system.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders
HTTP Operation	POST
Data Type	Order

- **orders/{id}/outage**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system creates or updates outage data in an order.
Description	When an order is created or updated in ADMS, Mobility Gateway sends order details to the Mobile Enterprise system. The update may include properties of the outage, details of the restoration order, and details of the incident. Mobile Enterprise updates the trouble order within the Mobile Enterprise system and on all field crew mobile devices for crews assigned to the order.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders/{id}/outage
HTTP Operation	POST, PATCH

Data Type (POST)	Outage
Data Type (PATCH)	OutageUpdate

- **orders/{id}/assignments**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system creates an assignment for an order.
Description	When a dispatcher assigns a crew to an order or updates order assignment, Mobility Gateway sends the order assignment details to the Mobile Enterprise system. Mobile Enterprise updates the order within the Mobile Enterprise system.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders/{id}/assignments
HTTP Operation	POST, PATCH
Data Type	OrderAssignment (list)

- **orders/{id}/switching**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system updates switching order that is associated with an order.
Description	When a switching order that is associated with an order is created in ADMS, Mobility Gateway sends the switching order details to the Mobile Enterprise system. Mobile Enterprise updates the order within the Mobile Enterprise system and on all field crew mobile devices for all crews assigned to the order.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders/{id}/switching
HTTP Operation	POST
Data Type	Switching

- **orders/{id}/switching/steps**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system creates or updates switching steps for an order.
Description	When a switching step that is associated with an order is created or updated in ADMS, Mobility Gateway sends the switching step details to the Mobile Enterprise system. Mobile Enterprise updates the order within the Mobile Enterprise system and on all field crew mobile devices for all crews assigned to the order.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders/{id}/switching/steps
HTTP Operation	POST, PATCH
Data Type (POST)	InsertSteps
Data Type (PATCH)	ModifySteps

- **orders/{id}/switching/order**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system updates a switching order for an order.
Description	When a switching order that is associated with an order is updated in ADMS, Mobility Gateway sends the switching order details to the Mobile Enterprise system. Mobile Enterprise updates the order within the Mobile Enterprise system and on all field crew mobile devices for all crews assigned to the order.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders/{id}/switching/order
HTTP Operation	PATCH
Data Type	SwitchingOrder

- **orders/{id}/outage/order/tasks**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system inserts a task into an order.
Description	When a task is created for an order in ADMS, Mobility Gateway sends the task details to the Mobile Enterprise system. Mobile Enterprise adds the task to the order within the Mobile Enterprise system and on all field crew mobile devices for all crews assigned to the order.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders/{id}/outage/order/tasks
HTTP Operation	POST
Data Type	Task

- **orders/{id}/outage/order/tasks/{taskId}**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system updates a task for an order.
Description	When a task that is associated with an order is updated in ADMS, Mobility Gateway sends the task details to the Mobile Enterprise system. Mobile Enterprise updates the order within the Mobile Enterprise system and on all field crew mobile devices for all crews assigned to the order.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders/{id}/outage/order/tasks/{taskId}
HTTP Operation	PUT
Data Type	Task

- **orders/{id}/names**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system updates order names.

Description	When an order name is updated in ADMS, Mobility Gateway sends the name update details to the Mobile Enterprise system. Mobile Enterprise updates the order within the Mobile Enterprise system.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders/{id}/names
HTTP Operation	PUT
Data Type	Name

- **orders/{id}/forms**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The ADMS system requests additional forms for an order.
Description	Mobility Gateway uses this message to request additional forms for an order to capture additional completion details due to the nature of the order.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/orders/{id}/assignments
HTTP Operation	GET
Data Type	Form

- **messages**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The ADMS system requests outbound messages or updates the message status.
Description	The ADMS system requests outbound messages from Mobile Enterprise or updates the message status.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/messages
HTTP Operation	GET, POST
Data Type (GET)	OutboundMessage
Data Type (POST)	OutboundMessageStatusUpdate

- **configuration/codedTranslations**

Host	Mobile Enterprise
Source	Mobility Gateway
Summary	The Mobile Enterprise system updates Coded Translations.
Description	Mobility Gateway uses this message to send Coded Translations to Mobile Enterprise system.
Default URI	http://localhost:8445/api/ext/MobileEnterprise/v1/configuration/codedTranslations
HTTP Operation	PUT
Data Type (PUT)	CodedTranslation (list)

3.3.3. Messages Web Services Hosted by Mobile Enterprise (Outbound/Inbound)

The following are default Mobile Enterprise web interface URLs that the Mobility Gateway communicates with. Mobile Enterprise queues messages as json objects and makes them ready for Mobility Gateway to consume filtered by date and time. Mobility Gateway could be configured to Post a message status for every message it gets from Mobile Enterprise. Each of the outbound/inbound messages has its own message type and schema. For more details about the data type format (message schema), see the mobile-enterprise-order-web-service.json Swagger definition file (recommended for viewing using <https://editor.swagger.io/>) and the mobile-enterprise-order-web-service.html file. These files are delivered with the ADMS documentation.

- **/api/ext/MobileEnterprise/v1/messages:** This is the root of the Mobile Enterprise messages web service.

All Mobile Enterprise messages, but OutboundMessageStatusUpdate, are outbound:

- **NotifyFieldDamageAssessment**

This message is used when Mobile Enterprise is set for Field Damage Assessment.

Note

Adding an "AssessmentLocation" that is not associated with any incident/order will not be displayed on Geographic view unless its "status" is "Not Repaired". Also, it is recommended to include assessment location coordinates. Otherwise, it will be very hard to locate an assessment location with (0, 0) coordinates.

▪ Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="NotifyFieldDamageAssessment" nillable="true"
type="NotifyFieldDamageAssessment" />
  <xs:complexType name="NotifyFieldDamageAssessment">
    <xs:complexContent mixed="false">
      <xs:extension base="OutboundMessageBody">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="AssessmentLocations"
type="ArrayOfAssessmentLocation" />
          <xs:element minOccurs="1" maxOccurs="1" name="DamagesProcess"
type="HttpMethodType" />
          <xs:element minOccurs="0" maxOccurs="1" name="Damages" type="ArrayOfDamage" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

<xs:complexType name="OutboundMessageBody">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="MessageType" type="MessageType" />
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="MessageType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="NotifyOrderAssignmentStatusEnum" />
    <xs:enumeration value="NotifySwitchingStepStatusEnum" />
    <xs:enumeration value="NotifySwitchingActionEnum" />
    <xs:enumeration value="NotifyCrewStatusEnum" />
    <xs:enumeration value="NotifyCrewLocationEnum" />
    <xs:enumeration value="NotifyOutageStatusEnum" />
    <xs:enumeration value="NotifyTaskStatusEnum" />
    <xs:enumeration value="NotifyUrgentWorkEnum" />
    <xs:enumeration value="NotifyLogActionEnum" />
    <xs:enumeration value="RequestCreateOrderFromFieldEnum" />
    <xs:enumeration value="NotifyFormEnum" />
    <xs:enumeration value="NotifyFieldDamageAssessmentEnum" />
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="ArrayOfAssessmentLocation">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="AssessmentLocation"
nillable="true" type="AssessmentLocation" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="AssessmentLocation">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="GeoLocation" type="GeoLocation" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="GeoLocation">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="ProjectionType" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="X" nillable="true" type="xs:double" />
    <xs:element minOccurs="1" maxOccurs="1" name="Y" nillable="true" type="xs:double" />
    <xs:element minOccurs="1" maxOccurs="1" name="Zone" nillable="true" type="xs:int" />
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="HttpMethodType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="POSTEnum" />
    <xs:enumeration value="PUTEnum" />
    <xs:enumeration value="DELETEEnum" />
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="ArrayOfDamage">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="Damage" nillable="true"
type="Damage" />
  </xs:sequence>
</xs:complexType>

```

```

    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="Damage">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="DamageId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="DamageAssessmentLocationId"
type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="AssetType" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="AssetSubType" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="Quantity" type="xs:int" />
      <xs:element minOccurs="0" maxOccurs="1" name="ActionType" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

▪ json-sample

```

{
  "Header": {
    "Date": "Tue, 04 Jun 2024 19:04:26 GMT",
    "Server": "Microsoft-HTTPAPI/2.0",
    "Content-Length": "785",
    "Content-Type": "application/json; charset=utf-8"
  },
  "Body": [
    {
      "header": {
        "messageId": "d73a1fa5-6b4e-4881-9d39-9394f40c0010",
        "messageTime": "2024-06-04T12:04:07.9769255-07:00"
      },
      "body": {
        "$type":
"OmsUtilities.InterfaceModels_V2.MobileEnterprise.NotifyFieldDamageAssessment,
OmsUtilities",
        "id": "ed818c57-9445-4340-b8f0-fad00b90e408",
        "incidentName": "1560000001",
        "crewId": "FI1943",
        "assessmentLocations": [
          {
            "id": "1e20b9ce-1286-41ea-8075-f294975ade39",
            "status": 0,
            "geoLocation": {
              "projectionType": "EPSG:4326",
              "x": -80.1576708938798,
              "y": 26.5600073017552,
              "zone": 0
            }
          }
        ],
        "damagesProcess": "post",
        "damages": [
          {
            "damageId": "1cf7ec56-826b-4ffb-b668-649ca45bc6d9",

```

```

        "damageAssessmentLocationId": "1e20b9ce-1286-41ea-8075-f294975ade39",
        "assetType": "Wires",
        "assetSubType": "Size 3/0, Aluminium",
        "quantity": 3,
        "actionType": "Repair"
      }
    ],
    "messageType": "NotifyFieldDamageAssessment"
  }
}
]
}

```

◦ **NotifyCrewAvailability**

When a crew updates his availability status, Mobile Enterprise (ME) creates this notify message and makes it available for Mobility Gateway to get.

i Note

ME gets all Coded Translation entries when Mobility Gateway starts. ME uses Coded Translation CrewAvailabilityStatus entry when sending this message to Mobility Gateway.

▪ **Schema**

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="NotifyCrewAvailability" nillable="true" type="NotifyCrewAvailability" />
  <xs:complexType name="NotifyCrewAvailability">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="UserId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="RecordedTime" nillable="true"
type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="SystemTime" nillable="true"
type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AvailabilityStatusCode"
type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="AvailabilityStatusText"
type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

▪ **json-sample**

```

{
  "Header": {
    "Date": "Wed, 06 Nov 2024 17:25:21 GMT",

```

```

    "Server": "Microsoft-HTTPAPI/2.0",
    "Content-Length": "386",
    "Content-Type": "application/json; charset=utf-8"
  },
  "Body": [
    {
      "header": {
        "messageId": "44567b19-7e30-4e9f-b7fa-76980a061955",
        "messageTime": "2024-11-06T09:25:17.6729282-08:00"
      },
      "body": {
        "$type": "OmsUtilities.InterfaceModels_V2.MobileEnterprise.NotifyCrewAvailability, OmsUtilities",
        "crewId": "FI1405",
        "userId": "FI1405",
        "deviceId": "",
        "availabilityStatusCode": "5",
        "availabilityStatusText": "Waiting On Equipment",
        "messageType": "NotifyCrewAvailability"
      }
    }
  ]
}

```

◦ **NotifyCrewStatus**

Mobile Enterprise (ME) sends this message to Mobility Gateway (MG) as the crew starts/ends shift, or goes offline. Regardless of order type, MG will:

- Update crew shift status based on configured maps (CrewShiftStatusMap) between ADMS and ME.
- Update the crew online/comm statuses for Switching Orders.

i Note

When sending this notification to MG, ME utilizes StatusEnum in the schema below.

▪ **Schema**

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="NotifyCrewStatus" nillable="true" type="NotifyCrewStatus" />
  <xs:complexType name="NotifyCrewStatus">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="UserId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true"
type="StatusEnum" />
      <xs:element minOccurs="1" maxOccurs="1" name="StatusTime" nillable="true"
type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="SystemTime" nillable="true"

```

```

type="xs:dateTime" />
</xs:sequence>
</xs:complexType>
<xs:simpleType name="StatusEnum">
  <xs:restriction base="xs:string">
    <xs:enumeration value="AVAILABLEEnum" />
    <xs:enumeration value="UNAVAILABLEEnum" />
    <xs:enumeration value="OFFLINEEnum" />
  </xs:restriction>
</xs:simpleType>
</xs:schema>

```

▪ json-sample

```

{
  "Header": {
    "Date": "Thu, 07 Nov 2024 16:51:20 GMT",
    "Server": "Microsoft-HTTPAPI/2.0",
    "Content-Length": "370",
    "Content-Type": "application/json; charset=utf-8"
  },
  "Body": [
    {
      "header": {
        "messageId": "7d20e9f7-617d-4fc6-ab67-c0bc91f05196",
        "messageTime": "2024-11-07T08:51:16.2605495-08:00"
      },
      "body": {
        "$type": "OmsUtilities.InterfaceModels_V2.MobileEnterprise.NotifyCrewStatus, OmsUtilities",
        "crewId": "FI1944",
        "userId": "FI1944",
        "deviceId": "YD-1234",
        "status": "AVAILABLE",
        "systemTime": "2024-11-07T08:50:42.4731798-08:00",
        "messageType": "NotifyCrewStatus"
      }
    }
  ]
}

```

◦ OutboundMessageStatusUpdate

This is a Mobile Enterprise inbound message. Mobility Gateway must have "EnableMobileEnterpriseMessageStatusUpdates" flag set to true. For each message Mobility Gateway gets from Mobile Enterprise, Mobility Gateway will Post an OutboundMessageStatusUpdate message to Mobile Enterprise.

▪ Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">

```

```

<xs:import namespace="http://microsoft.com/wsd1/types/" />
<xs:element name="OutboundMessageStatusUpdate" nillable="true"
type="OutboundMessageStatusUpdate" />
<xs:complexType name="OutboundMessageStatusUpdate">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="MessageId" nillable="true"
xmlns:q1="http://microsoft.com/wsd1/types/" type="q1:guid" />
    <xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true"
type="StatusEnum" />
    <xs:element minOccurs="0" maxOccurs="1" name="Reason" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="StatusEnum">
  <xs:restriction base="xs:string">
    <xs:enumeration value="SUCCESSEnum" />
    <xs:enumeration value="RETRYEnum" />
    <xs:enumeration value="FAILEDEnum" />
  </xs:restriction>
</xs:simpleType>
</xs:schema>

```

▪ json-sample

```

{
  "Header": {
    "Accept": "application/json",
    "Content-Type": "application/json"
  },
  "Body": {
    "messageId": "d73a1fa5-6b4e-4881-9d39-9394f40c0010",
    "status": "SUCCESS",
    "reason": "NotifyFieldDamageAssessment- Order:1560000001 Crew:FI1943.
CrewServerClient: SUCCESSEnum "
  }
}

```

4. Optional Features

This chapter describes the optional features that Mobility Gateway provides based on configuration.

4.1. Field Damage Assessment

Mobility Gateway can be configured to notify Mobile Enterprise about an order that requires assessment. Mobility Gateway can also handle `asset-type-based` damage assessment messages from Mobile Enterprise, whether the message specifies an associated order or not. Such a message creates appropriate damage assessment locations and associated damage(s), and associates them with the specified order (if provided).

4.2. Order Suspension

Mobility Gateway can automatically suspend and unsuspend an order without operator intervention. With appropriate configuration, order suspension and unsuspension are applied based on crew status changes initiated by Mobile Enterprise. When automatic suspension is carried out, the suspension time is set and the suspension reason can be set to a configured value.

- **Auto Suspension:** An order is suspended when Mobile Enterprise notifies Mobility Gateway that a Dispatched, EnRoute or Arrived (Started) crew has Rejected an order. Mobility Gateway suspends the order with configured reason, provided the `"SuspendOrderOnCrewRelease"` and `"ReleasedSuspensionReason"` are correctly set.

Note

Only the configured suspension reason is used.

- **Manual Suspension:** An order is suspended when Mobile Enterprise notifies Mobility Gateway that an EnRoute or Started crew has put their work OnHold or Suspended. Mobility Gateway suspends the order with configured reason, provided the `"SuspendOrderOnCrewSuspension"` and `"StoppedSuspensionReason"` are correctly set.

Note

If the notification message includes a valid reason from the `"WorkOrderSuspendedReasons"` coded translation, that reason is used instead of the configured default.

- **Unsuspend Order:** Mobility Gateway changes a `"Suspended"` order to `"InProgress"` when the crew status is changed to `"ENROUTE"` or `"ARRIVED"` by Mobile Enterprise. The `"CrewStatusesToUnsuspendOrder"` must be set to `"ENROUTE"` and/or `"ARRIVED"`.

5. Configuring the System

This chapter describes changes that must be made to the default demo (Fantasy Island) configurations to enable integration between ADMS and Mobile Enterprise using the Mobility Gateway application. For more details about the required configurations, refer to the appendix "Configuring Mobile Dispatching" in the *Switching Order Server Configuration Editor User Guide*.

Configuring the Switching Order Server

Configure the `SwitchOrderServerConfig.txt` file (by default, stored at `%IDMS_ROOT%\config\server`). For more information, refer to the *Switching Order Server Configuration Editor User Guide*.

Set the configurations as follows:

```
MobilePlugin = MobilePlugin.dll
EmployeeStatusFile = %IDMS_ROOT%\dynamics\server\SwitchOrderServer\LogonStatus.xml
DispatchStatusFile = %IDMS_ROOT%\dynamics\server\SwitchOrderServer\DispatchStatus.xml
MobileEmployeeList = %IDMS_ROOT%\config\server\MobilePluginEmployees.txt
CrewList = %IDMS_ROOT%\models\crew\FantasyIslandCrewModel.Crew
MobilityGatewaySpecific = true
ExpectMobileUpdateAck = true
```

Configuring the Switching Order Client

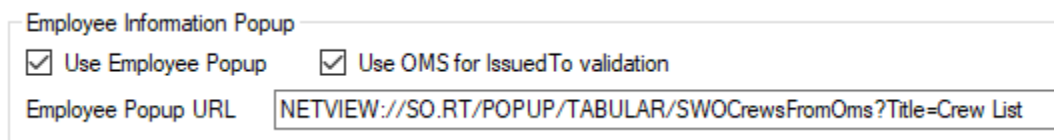
Configure the `SwitchOrderClientConfig.xml` file (by default, stored at `%IDMS_ROOT%\config\client`). For more information, refer to the *Switching Order Client Configuration Editor User Guide*.

Set the configurations as follows:

```
<bUseOmsAsCrewSource>true</bUseOmsAsCrewSource>
<SwoEmployeePopupUrl>NETVIEW://SO.RT/POPUP/TABULAR/SWOCrewsFromOms?Title=Crew
List</SwoEmployeePopupUrl>
```

Alternatively, you may perform these configurations using the Switch Order Client Configuration Editor.

1. Open the Switch Order Client Configuration Editor.
2. Open the Switch Order related tab.
3. Select the "Use OMS for IssuedTo Validation" option.
4. Set the "Employee Popup URL" value to "NETVIEW://SO.RT/POPUP/TABULAR/SWOCrewsFromOms?Title=Crew List"



Employee Information Popup

☒ Use Employee Popup ☒ Use OMS for IssuedTo validation

Employee Popup URL NETVIEW://SO.RT/POPUP/TABULAR/SWOCrewsFromOms?Title=Crew List

5. Select the Enable Document Dispatching option.

Mobile Options

☒ Enable document dispatching

Configuring the Crew Status Summary Grid Tabular Display

1. Open the `CrewStatusGridDisplay.xml` file (by default, stored at `%IDMS_ROOT%\tabulars`) in a text editor.
2. Search for the following string:

```
<CellName>CREWORKSTATUS %CREWORKSTATUSCODE</CellName>
```

3. Change the `HideByDefault` value to `false`.

```
<HideByDefault>>false</HideByDefault>
```

This optional change shows the Crew Working Status column on the Crew Status Grid display in the OMS Display plugin. The Mobility Gateway updates the Crew Working Status field with the crew status that comes from Mobile Enterprise.

6. Configuration Files

6.1. Mobility Gateway Configuration

This section highlights some important Mobility Gateway configuration settings but all of them are not documented here. For more details about different options and configurations, please refer to the `MobilityGatewayAppConfig.xml` configuration file.

The Mobility Gateway application reads the `MobilityGatewayAppConfig.xml` configuration file, which is located in the `%IDMS_ROOT%/config/interfaces` folder.

The `MobilityGatewayAppConfig.xml` configuration file includes the following parameters:

- **EnableDebugLog:** Specifies whether to log detailed information for debugging purposes.
- **LogFile:** Specifies the full path to the file that is used for Mobility Gateway logging purposes.
- **MaxMessagesInLogFile:** The maximum number of records in the log file. When this number is reached, another log file is created. The default value is 10,000.
- **Log Service Plugin Parameters:** This group includes parameters for the Mobility Gateway logging service plugin.
 - **Enabled:** If set to true, the Mobility Gateway logging service plugin is enabled.
 - **Type:** The type that implements the Mobility Gateway logging service plugin.
 - **Name:** Mobility Gateway logging service plugin name.
 - **LogMessageTypes:**
 - **Type:** Application for which the log is created.
 - **SeparateLogToDisk:** If set to true, a separate log is created for this type.
 - **LogToDisk:**
 - **Name:** Name of the log file.
 - **EnableDailyLog:** If set to true, a separate log file is created daily.
 - **MaxEntries:** Specifies the maximum number of entries in the log file.
- **OMS Interface Server Query Web Service Plugin Parameters:** This group includes OMS Interface Server Query web service plugin parameters. This plugin supports general and predefined queries to the OMS Interface Server's Query web service.
 - **Enabled:** If set to true, the OMS Interface Server Query web service plugin is enabled.
 - **Type:** The type that implements the OMS Interface Server Query web service plugin.
 - **Name:** OMS Interface Server Query web service plugin name.
 - **BasePath:** OMS Interface Server Query web service root URL.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version

parameters.

- **Name:** OMS Interface Server web service name to be shown on the user interface.
- **OmsQueryUri:** The endpoint for OMS queries.
- **IncidentUri:** The endpoint for incident queries.
- **PlannedOutageUri:** The endpoint for planned outage queries.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **OMS Interface Server V1 Web Service Client Plugin Parameters:** This group includes OMS Interface Server V1 web service client plugin parameters. This plugin interacts with the Oms Interface Server's REST web service, which uses the "oms/services/rest" endpoint.
 - **Enabled:** If set to true, the OMS Interface Server V1 web service client plugin is enabled.
 - **Type:** The type that implements the OMS Interface Server V1 web service client plugin.
 - **Name:** OMS Interface Server V1 web service client plugin name.
 - **Config > Name:** OMS Interface Server V1 web service client to be shown on the user interface.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **UpdateAreaNoteUri:** The endpoint for update area settings queries.
 - **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier (a string date and time value with time zone information). For more information, see [Time Format](#).
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **Work Model Client Plugin Parameters:** This group includes Work Model client plugin parameters. This interface defines the methods that support order and task processing in the Enterprise OMS system via the Mobility Gateway.
 - **Enabled:** If set to true, the Work Model client plugin is enabled.
 - **Type:** The type that implements the Work Model client plugin.
 - **Name:** Work Model client plugin name.
 - **Config > Name:** Work Model client name to be shown on the user interface.
 - **Description:** Work Model client description.
 - **WorkOrderServiceUrl:** Enterprise Work Model Event Handler REST API endpoint.
 - **WorkOrderTlsEnabled:** Specifies whether the HTTPS TLS security is enabled for the Work Model Event Handler service connection for web clients and web socket clients.
 - **WorkOrderTlsSubjectName:** The certificate subject name that is used when searching for

client certificates to send to the Work Model Event Handler service.

- **WorkModelTlsIgnoreRemoteNameMismatch:** Specifies whether the remote server certificate name mismatch is ignored. If set to True, the remote certificate name can be different from the URL host name that is used to connect to the server.
- **Mobile Enterprise Web Service Client Plugin Parameters:** This group includes Mobile Enterprise web service client plugin parameters. This plugin implements the client-side logic, model transformations, and HTTP POST and GET methods for interacting with Mobile Enterprise.
 - **Enabled:** If set to true, the Mobile Enterprise web service client plugin is enabled.
 - **Type:** The type that implements the Mobile Enterprise web service client plugin.
 - **Name:** Mobile Enterprise web service client plugin name.
 - **Config > Name:** Mobile Enterprise web service client name to be shown on the user interface.
 - **Description:** Mobile Enterprise web service client description.
 - **BaseUri:** Mobile Enterprise web service client root URL.
 - **OrdersUri:** Orders endpoint.
 - **MessagesUri:** Outbound messages endpoint.
 - **MessageGetPollingInterval:** Polling time for Mobile Enterprise web service client messages, in seconds.
 - **CrewSynchronizationUri:** Crew Synchronization endpoint.
 - **CodedTranslationsUri:** Coded Translations endpoint.
 - **CodedTranslationsXMLPath:** Coded Translations XML configuration file path.
 - **TagsUri:** The URI for Posting Tags to Mobile Enterprise endpoint (for example, Yambay, <https://%2e%2e%2e/aov/tags>)
 - **TagQueryPollingInterval:** Polling interval (in seconds) to query for Tags from the Dms Interface Server. Set to 0 to disable. The default is 120 seconds (2 mins).
 - **PersistentLastDataFile:** The persistent data file records the timestamp of the last processed message ("from") so that the already processed messages are not processed again if the Mobile Enterprise web service client is restarted.
 - **PersistentLastSwoEventSequenceFile:** Stores the sequence number of the Switching Order event sent to Mobile Enterprise.
 - **PersistentLastWorkModelEventSequenceFile:** Stores the sequence number of the last WorkModel event sent to Mobile Enterprise.
 - **Plugins:** List of plugins the Mobile Enterprise web service client uses:
 - WorkModelQueryService
 - OisV2QueryService
 - OisV1QueryService

- DmsQueryService
- TaggingQueryService
- CrewServerWebService
- DisSwoWebService
- **RouteOrderAssignmentStatusCrewServer:** If set to true, order crew status changes are also sent to the Crew Server to be available in the Work Status field on the Crews tab in the OMS Display plugin.
- **FieldDamageAssessmentOnly:** If set to true, Mobility Gateway will only process EOMS "Task-Added" events for "Assessment" task only. Mobility Gateway will compile an "outage" payload and post it to Mobile Enterprise. For efficiency, except for "Task", all WorkOrderEvents "Enabled" flag should be configured "false". May also filter the "added" tasks only.
- **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier (a string date and time value with time zone information). For more information, see [Time Format](#).
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
- **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
- **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.
- **MaxPostTryCnt:** The number of times to try to deliver a message. If this count is reached for a message, the message is rejected and eliminated from the retry logic. To disable limiting the number of retries, set this parameter to "-1".
- **HttpRequestTimeoutSeconds:** Time period to wait for the client response for an HTTP request. This parameter is used for testing and debug purposes.
- **HttpDefaultRequestHeadersExpectContinue:** Specifies whether the HttpClient "ExpectContinue" behavior is enabled.

By default, DotNet enables the "ExpectContinue" behavior. This behavior results in HTTP

clients posting the "Expect: 100-Continue" request header first, and then waiting for a "100-Continue" response from the server before posting the request payload. Disable this functionality when using load balancer software that has issues supporting this functionality. Disabling this functionality eliminates one extra client server exchange for each message posted.

- **ProjectionType:** Projection type (for example, GEO).
- **ProjectionZone:** Universal Transverse Mercator (UTM) coordinate system projection zone.

Note

To use system coordinates (X, Y) for the geolocation, populate the `ProjectionType` and `ProjectionZone` values and comment out the `CoordinateConverterClass` and `CoordinateConverterDll` parameters.

- **CoordinateConverterClass:** Name of the class that converts system coordinates (X, Y) into projected coordinate system (PCS) coordinates (latitude, longitude). The value format is `namespace.class` (for example, `SampleCoordinateConverter.CoordinateConverter`).
- **CoordinateConverterDll:** Name of the assembly (DLL file) that contains a class for converting system X/Y coordinates into latitude and longitude coordinates. The default assembly, `NetUtil.dll`, is located in the `/server/bin` folder.

Note

For US customers, the default coordinate converter can be used. For non-US customers, a custom converter must be developed.

- **SuspendOrderOnCrewRelease:** Controls whether to suspend an order when its last crew is released or unassigned. Should be true to enable or false to disable.
- **ReleasedSuspensionReason:** The value to be used for an order's `SuspensionReason` when it is automatically suspended by Mobility Gateway due to the unassignment or release of the last crew. Should be a valid key from the 'WorkOrderSuspendedReasons' coded translation.
- **SuspendOrderOnCrewSuspension:** Controls whether to suspend an order when an individual crew is suspended or put on hold. Should be true to enable or false to disable.
- **StoppedSuspensionReason:** The value to be used for an order's `SuspensionReason` when Mobility Gateway suspends it due to an explicit suspension request and if the request doesn't also include a valid suspension reason. Should be a valid key from the 'WorkOrderSuspendedReasons' coded translation.
- **CrewStatusesToUnSuspendOrder:** The crew statuses (if any) that, when being set by Mobility Gateway, should unsuspend a suspended order. Valid values are `ENROUTE` and `ARRIVED` or both (comma separated).
- **CrewShiftStatusMap:** Maps each of the Mobile Enterprise Crew Status values to the corresponding key of Coded Translation `CrewShiftStatus`.

- **MeCrewStatus:** Name of the crew status coming from Mobile Enterprise (AVAILABLE, UNAVAILABLE or OFFLINE).
- **AdmsShiftStatus:** Key of the Coded Translation CrewShiftStatus to be mapped with MeCrewStatus above.
- **Order Events Processor Plugin Parameters:** This group includes Work Order Events Processor plugin parameters. This plugin listens to the Work Model service and Work Order database update event stream and forwards the events to the Mobile Enterprise web service client plugin for forwarding to Mobile Enterprise.
 - **Enabled:** If set to true, the Work Order Events Processor plugin is enabled.
 - **Type:** The type that implements the Work Order Events Processor plugin.
 - **Name:** Order Events Processor plugin name.
 - **PrereqPlugins:** Comma-separated list of plugins that the Work Order Events Processor plugin uses. The required plugins must be loaded and available before the Order Events Processor plugin is loaded.
 - **Config > Name:** Work Order Events Processor name to be shown on the user interface.
 - **Description:** Work Order Events Processor description.
 - **WorkOrderEventsWebSocketUri:** Work Model Event Handler Service web socket client URL.
 - **MobileWfmWebServiceClient:** Mobile Enterprise web service client root URL.
 - **WorkOrderEvents:** Specifies order events that are processed by the Work Order Events Processor plugin and made ready to be processed by Mobility Gateway.
 - **Name:** Name of the event (order, task, link, or crew).
 - **Enabled:** Specifies whether to process the event or not.
 - **Filter:** The list of action types that are processed (added, updated, or deleted).
- **Mobile Switching Order Web Service Plugin Parameters:** This group includes Mobile Switching Order web service plugin parameters. The Mobile Switching Order web service plugin implements a web service to which the DMS Interface Server sends switching order events.
 - **Enabled:** If set to true, the Mobile Switching Order web service plugin is enabled.
 - **Type:** The type that implements the Mobile Switching Order web service plugin.
 - **Name:** Mobile Switching Order web service plugin name.
 - **Config > Name:** Mobile Switching Order web service name to be shown on the user interface.
 - **Description:** Mobile Switching Order web service description.
 - **BaseUri:** Mobile web service client root URL.
 - **Plugins:** List of plugins the Mobile Switching Order web service uses:
 - **MobileSWOWebServiceClient (MobileEnterpriseWebServiceClient):** The Mobile

Switching Order web service uses the Mobile Enterprise web service client, which provides methods to interact with the REST service in the Mobile Enterprise system.

- **EnableAuthentication:** Set to true to enable authentication.
- **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **DIS SWO Web Service Client Plugin Parameters:** This group includes DMS Interface Server Switching Order (DIS SWO) web service client plugin parameters. This plugin implements the client-side logic, model transformations, and HTTP POST and GET methods for interacting with the DMS Interface Server's Switching Order web services.
 - **Enabled:** If set to true, the DIS SWO web service client plugin is enabled.
 - **Type:** The type that implements the DIS SWO web service client plugin.
 - **Name:** DIS SWO web service client plugin name.
 - **Config > Name:** DIS SWO web service client name to be shown on the user interface.
 - **BaseUri:** DIS SWO web service client root URL.
 - **MessagesUri:** Messages endpoint.
 - **MockData:** Specifies whether mock data is used (a set of test data is returned without a connection to the DMS Interface Server). This parameter is used for testing purposes.
 - **PostingRetryInterval:** Time period, in seconds, to retry posting messages to the remote endpoint when posting failed (for example, due to connection issues).
 - **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier (a string date and time value with time zone information). For more information, see [Time Format](#).
 - **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
 - **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
 - **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
 - **CertSubjectName:** The TLS certificate subject name's CN value.
 - **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
 - **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
 - **UaaEndPoint:** Endpoint for communicating with the UAA server and retrieving a valid UAA authentication token.

- **BlockPostMessageTypes:** DMS Interface Server service message types to block from being sent to the DIS SWO web service client (for example, Status).
- **RejectedHttpResponseStatusCodes:** HTTP response codes that cause the associated message to be rejected and eliminated from the retry logic.
- **MaxPostTryCnt:** The number of times to try to deliver a message. If this count is reached for a message, the message is rejected and eliminated from the retry logic. To disable limiting the number of retries, set this parameter to "-1".
- **HttpRequestTimeoutSeconds:** Time period to wait for the client response for an HTTP request. This parameter is used for testing and debug purposes.
- **HttpDefaultRequestHeadersExpectContinue:** Specifies whether the HttpClient "ExpectContinue" behavior is enabled.

By default, DotNet enables the "ExpectContinue" behavior. This behavior results in HTTP clients posting the "Expect: 100-Continue" request header first, and then waiting for a "100-Continue" response from the server before posting the request payload. Disable this functionality when using load balancer software that has issues supporting this functionality. Disabling this functionality eliminates one extra client server exchange for each message posted.

- **HttpKeepAlive:** Specifies whether the HttpClient "KeepAlive" behavior is enabled.

By default, DotNet enables the "KeepAlive" behavior. This behavior specifies that HTTP clients maintain the existing HTTP connection when sending messages. When this functionality is disabled, every message is sent via a new HTTP connection.

- **Crew Server Web Service Client Plugin Parameters:** This group includes Crew Server web service client plugin parameters. This plugin implements the client-side logic, model transformations, and HTTP POST and GET methods for interacting with the Crew Server's Crew web services.
 - **Enabled:** If set to true, the Crew Server web service client plugin is enabled.
 - **Type:** The type that implements the Crew Server web service client plugin.
 - **Name:** Crew Server web service client plugin name.
 - **RetryEnabled:** If set to false, the retry posting is disabled for all plugin endpoints. If set to true, the "Retry" flag at the level of each endpoint determines whether retries are enabled for the associated endpoint. (Retries may be posted if a certain HTTP response status code is returned.)
 - **VerboseLogging:** Specifies whether verbose logging is enabled.
 - **BaseUri:** Crew Server web service client root URL.
 - **Config > Name:** Crew Server web service client name to be shown on the user interface.
 - **Description:** Crew Server web service client description.
 - **DateTimeFormatString:** Date and time format. By default, "O", which is a custom "round-trip" DateTime format specifier (a string date and time value with time zone information).

For more information, see [Time Format](#).

- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.

6.2. Mobile Enterprise Simulator Configuration

This section highlights some important Mobility Enterprise Simulator configuration settings but all of them are not documented here. For more details about different options and configurations, please refer to the `MobileEnterpriseSimulatorConfig.xml` configuration file.

The Mobile Enterprise Simulator application reads the `MobileEnterpriseSimulatorConfig.xml` configuration file, which is located in the `%IDMS_ROOT%/config/interfaces` folder.

The `MobileEnterpriseSimulatorConfig.xml` configuration file includes the following parameters:

- **EnableDebugLog:** Specifies whether to log detailed information for debugging purposes.
- **LogFile:** Specifies the full path to the file that is used for Mobility Gateway logging purposes.
- **MaxMessagesInLogFile:** The maximum number of records in the log file. When this number is reached, another log file is created. The default value is 10,000.
- **Plugin API Parameters:** This group includes base URI parameters for Mobile Enterprise Simulator plugins.
 - **UriScheme:** URI scheme: HTTP or HTTPS.
 - **UriHost:** URI host.
 - **UriPort:** URI port.
 - **UriPath:** URI path.
 - **FullMessages:** If set to true, entire message payloads are written to the log file.
- **Log Service Plugin Parameters:** This group includes parameters for the Mobile Enterprise Simulator logging service plugin.
 - **Enabled:** If set to true, the Mobile Enterprise Simulator logging service plugin is enabled.
 - **Type:** The type that implements the Mobile Enterprise Simulator logging service plugin.
 - **Name:** Mobile Enterprise Simulator logging service plugin name.

- **LogMessageTypes:**
 - **Type:** Application for which the log is created.
 - **SeparateLogToDisk:** If set to true, a separate log is created for this type.
 - **DisableLogToConsole:** If set to true, logging to the console is disabled for the associated plugins. Use this parameter to disable logging for certain plugins and declutter the application console log.
- **LogToDisk:**
 - **Name:** Name of the log file.
 - **EnableDailyLog:** If set to true, a separate log file is created daily.
 - **MaxEntries:** Specifies the maximum number of entries in the log file.
- **Mobile Enterprise Simulator Web Service Plugin Parameters:** This group includes Mobile Enterprise Simulator web service plugin parameters.
 - **Enabled:** If set to true, the Mobile Enterprise Simulator web service plugin is enabled.
 - **Type:** The type that implements the Mobile Enterprise Simulator web service plugin.
 - **Name:** Mobile Enterprise Simulator web service plugin name.
 - **Config > Name:** Mobile Enterprise Simulator web service name to be shown on the user interface.
 - **BasePath:** Mobile Enterprise Simulator web service root URL.
 - **MessagesUri:** Messages endpoint.
 - **OrdersUri:** Orders endpoint.
 - **Version:** Mobile Enterprise Simulator web service version.
 - **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.

7. Mobile Enterprise Simulator Testing Tool

The Mobile Enterprise Simulator application is designed for testing the Mobile Enterprise interface. Mobile Enterprise Simulator must not be used for production. The application hosts endpoints similar to the Mobile Enterprise system and can be used to test message format and endpoint interaction, and to simulate the types of messages that are retrieved from the Mobile Enterprise system. It is also possible to use Postman or a similar tool for API testing.

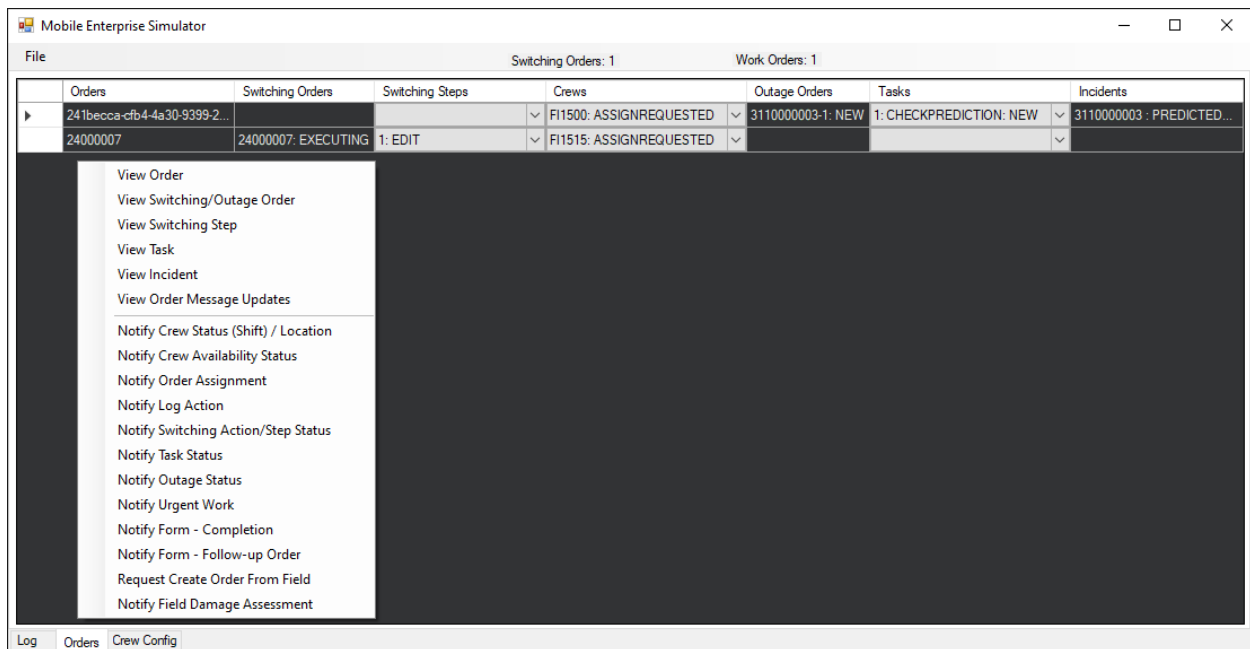


Figure 2. Mobile Enterprise Simulator - Orders

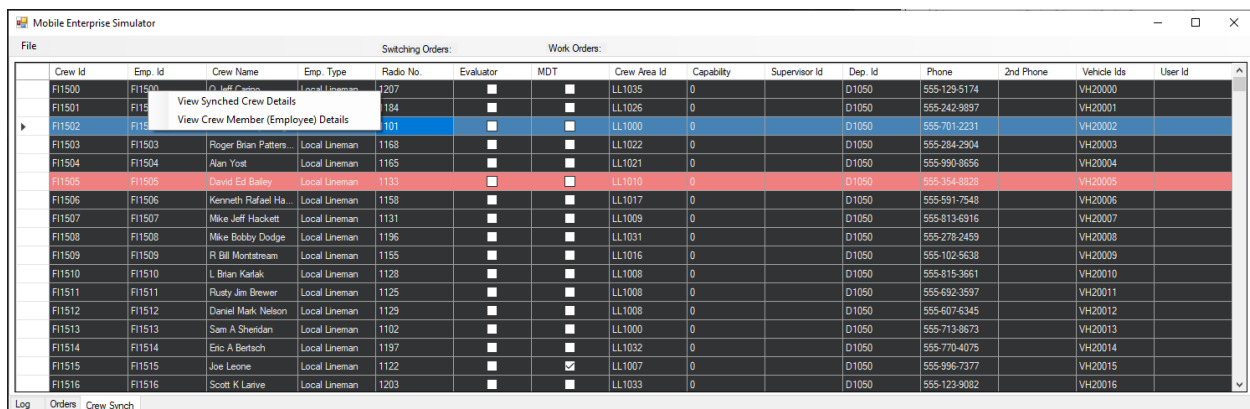


Figure 3. Mobile Enterprise Simulator - Synchronized Crews

To start the Mobile Enterprise Simulator, open the Windows Start menu and navigate to All Programs > Eterra > Distribution > Interfaces > Mobile Enterprise Simulator.

By default, the Mobile Enterprise Simulator starts with the MobileEnterpriseSimulatorConfig.xml configuration file (by default, stored at %IDMS_ROOT%\config\interfaces. For more details about the Mobile Enterprise Simulator configuration, see [Mobile Enterprise Simulator Configuration](#).

The Mobile Enterprise Simulator requires that the Mobility Gateway is running.